

LangChain PDF Loader and Virtual Environment Issue Report

Project

Building a RAG (Retrieval-Augmented Generation) application using LangChain and Jupyter Notebook.

Problem 1: Import Error

Error

```
from langchain.text_splitter import RecursiveCharacterTextSplitter
```

Result:

```
ModuleNotFoundError: No module named 'langchain.text_splitter'
```

Cause

The tutorial being followed used an older version of LangChain. In newer versions, the `RecursiveCharacterTextSplitter` class has been moved to a separate package.

Solution

Install the package:

```
pip install langchain-text-splitters
```

Use the updated import:

```
from langchain_text_splitters import RecursiveCharacterTextSplitter
```

Problem 2: PDF Loading Error

Error

```
`pypdf` package not found, please install it with `pip install pypdf`
```

Cause

PyPDFLoader depends on the pypdf library, but it was not installed in the active Python environment.

Solution

Install the package:

```
pip install pypdf
```

Problem 3: Unable to Import pypdf

Error

```
import pypdf
```

Result:

```
ModuleNotFoundError: No module named 'pypdf'
```

Cause

The package was installed in a different Python installation than the one used by the Jupyter notebook.

Investigation

Running:

```
import sys
print(sys.executable)
```

showed that the notebook was using:

```
D:\YTRAG\.venv\Scripts\python.exe
```

while packages had been installed in another Python installation.

Solution

Install packages inside the active virtual environment or switch the notebook to the correct interpreter.

Problem 4: pip Missing in Virtual Environment

Error

```
No module named pip
```

or

```
d:\YTRAG\.venv\Scripts\python.exe: No module named pip
```

Cause

The virtual environment was incomplete or corrupted and did not contain `pip`.

Solution

Recreate the virtual environment:

```
python -m venv .venv
```

Activate it:

```
.\.venv\Scripts\Activate.ps1
```

Upgrade pip:

```
python -m pip install --upgrade pip
```

Install required packages:

```
python -m pip install pypdf langchain langchain-community langchain-text-splitters pymupdf jupyter ipykernel
```

Problem 5: Virtual Environment Recreation Error

Error

```
Unable to copy venvlauncher.exe to .env\Scripts\python.exe
```

Cause

The existing virtual environment was still active or files were locked by VS Code or PowerShell.

Solution

1. Close VS Code.
2. Close all terminals.
3. Delete the existing `.env` folder.
4. Open a new terminal.
5. Run:

```
python -m venv .env
```

1. Activate the new environment and reinstall all dependencies.

Final Outcome

After recreating the virtual environment and installing all required packages into the correct interpreter:

- LangChain imports worked successfully.
- `RecursiveCharacterTextSplitter` imported correctly.

- `pypdf` was available.
 - `PyPDFLoader` loaded PDF files successfully.
 - The Jupyter notebook kernel used the correct Python environment.
 - The RAG project was ready for further development.
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Lessons Learned

- Always verify the active Python interpreter before installing packages.
- Create a dedicated virtual environment for every project.
- Install dependencies inside the active virtual environment.
- Older tutorials may use outdated imports that need updating for newer package versions.
- If a virtual environment is corrupted, recreating it is often the fastest and cleanest solution.